

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

`show ip route ospf`

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? s/0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? yes

Does R2 show up as an OSPF neighbor on R3? no

What does this information tell you?

All traffic from R3 destined for the 192.168.2.0/24 network will be forwarded through R1. Since the S0/0/1 interface on R2 is configured as a passive interface, OSPF updates are not sent over that link.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

`router ospf 1`

`no passive-interface s0/0/1`

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? s0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

Accumulated cost metric : 65. a T1 serial link(64) + R2 Gigabit 0/0 LANlink

Is R2 listed as an OSPF neighbor to R3? yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

% OSPF: Reference bandwidth is changed.

Please ensure reference bandwidth is consistent across all routers.

Why would you want to change the OSPF default reference-bandwidth?

To achieve a more accurate cost calculation, the reference bandwidth setting needs to be increased.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

192.168.3.0/24: pass R1 interface S0/0/1 and R3 interface G0/0, total 782

192.168.23.0/30: pass R1 interface S0/0/1 and R3 interface S0/0/1, total $781 + 64 = 845$

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

A single serial link has a cost of 781. Since the route to 192.168.23.0/24 traverses two serial links, the total cost is $2 \times 781 = 1562$.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

OSPF selects the route with the lowest total cost. In this case, the path with the lowest cumulative cost is $781 + 781 + 1 = 1,563$, which is lower than the alternative route through R1-S0/0/1 and R3-G0/0, whose total cost is 1,565.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

The router ID plays a crucial role in the election of the Designated Router and Backup Designated Router on multi-access networks. If it is based on an active physical interface, it may change when that interface goes down, which can cause instability in the OSPF process. To ensure consistency, the router ID should be assigned using the IP address of a loopback interface, since a loopback interface remains active.

2. Why is the DR/BDR election process not a concern in this lab?

In this lab, the serial links are configured as point-to-point connections, so a DR/BDR election is not required. The DR/BDR election process applies only to multi-access networks.

3. Why would you want to set an OSPF interface to passive?

This configuration reduces unnecessary OSPF traffic by suppressing Hello packet transmission on the interface. It also improves security by preventing unauthorized devices from forming OSPF neighbor adjacencies.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

The use of areas improves the scalability of the protocol, as routers within an area only need to maintain the topology information for that specific area.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

`network 10.0.0.0 0.255.255.255 area 0`

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

The EIGRP route with a metric of 4039043 will be chosen as the best path because EIGRP has the lowest administrative distance (90), compared to OSPF (110) and RIP (120). In cases where multiple routes have the same administrative distance, the router selects the one with the lowest metric.